EXECUTIVE SUMMARY

Behind Every Click, a Conversation: Listening to a Network and Knocking on Its Doors

- Every time a page loads, a video starts or a message is sent, a computer quietly holds dozens of conversations with servers around the world. Most of them go unnoticed.
- This document listens in. It captures live Wi-Fi traffic with Wireshark, filters out the noise, and reads the packets that reveal who is talking, what they ask for, and the moment something goes wrong.
- Then it does what every defender must do next: it checks the doors. With Nmap, it scans a computer and a few web servers to see which ports answer, which stay silent, and what each open port could mean for security.
- Together, the two tools show how an analyst moves from watching traffic to understanding exposure, one packet and one port at a time.

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The objective of this practical is to learn how to capture, filter and read network traffic with Wireshark, and how to scan ports and read the results with Nmap, while looking at every result from a defender point of view.

- Capture live traffic on a Wi-Fi connection and use display filters to find the traffic that matters.
- Read packet details and protocol layers, and understand TCP flags and retransmissions.
- Use Wireshark tools to summarise a capture: export, Follow TCP Stream, Expert Information, Conversations, Endpoints, name resolution and time display.
- Spot unusual behavior, such as the same connection attempt repeated again and again.
- Use Nmap through Zenmap to scan this computer and permitted targets, and read the states open, closed and filtered.
- Explain what each result means for a SOC analyst.

Lab details

Item	**Detail**
Tools	Wireshark (packet capture and analysis). Nmap 7.991 with its graphical program Zenmap (port scanning).
Scan scope	This computer, a web server whose owner gave permission, the university website, and scanme.nmap.org, which the Nmap project provides for scan practice.

PART A: WIRESHARK PRACTICAL

In this part, traffic was captured on the Wi-Fi interface of this computer. Each step gives its purpose, the action or filter used, the figure, and a defender view. Several captures were made, so the packet counts differ between some figures.

Capturing live traffic

Purpose: To record the packets that move through the Wi-Fi adapter while the computer is in use. The capture is the raw material for every other step in this document.

Action: Select the Wi-Fi interface and press the blue shark fin button (Start capturing packets). Press the red square button to stop the capture.

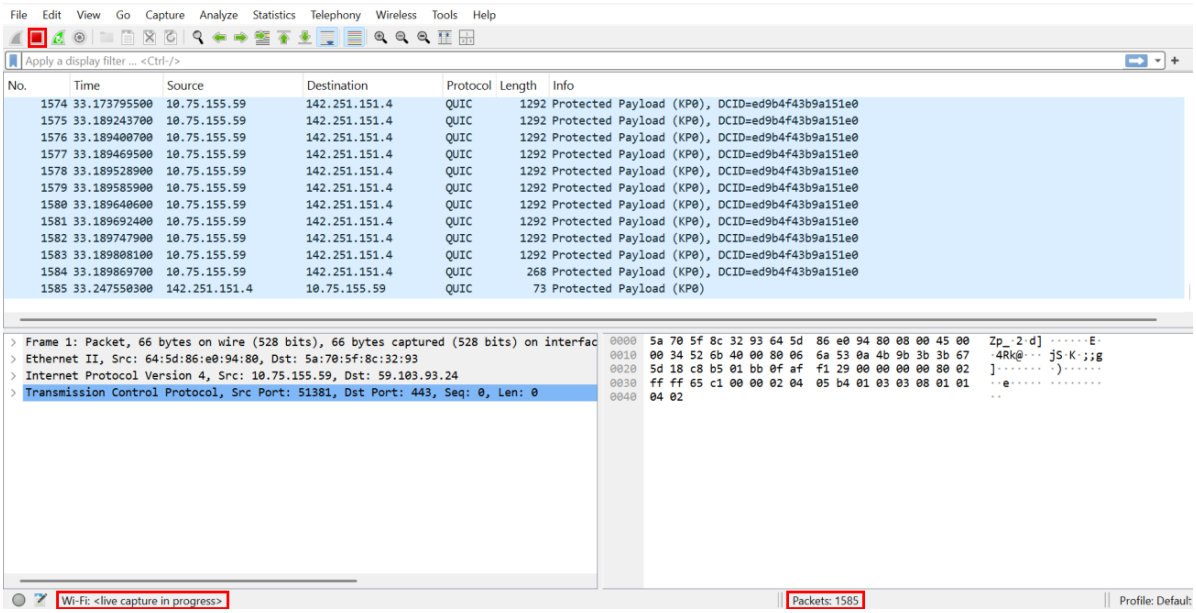


Figure 1: Live capture running on the Wi-Fi interface

Defender view: A capture is evidence. An analyst starts a capture on a suspicious computer, then uses filters to find the packets that matter. Because the payload here is encrypted, the analyst cannot read the content and must rely on details such as who talked to whom, when, and how much data moved.

Display filter for TCP

Purpose: To hide all packets except the ones that use TCP (Transmission Control Protocol), so that connections can be studied without noise.

Action: Type the filter below in the display filter bar and press Enter.

tcp

tcp: show only packets that use the TCP protocol. Packets that use UDP, ICMP and other protocols are hidden.

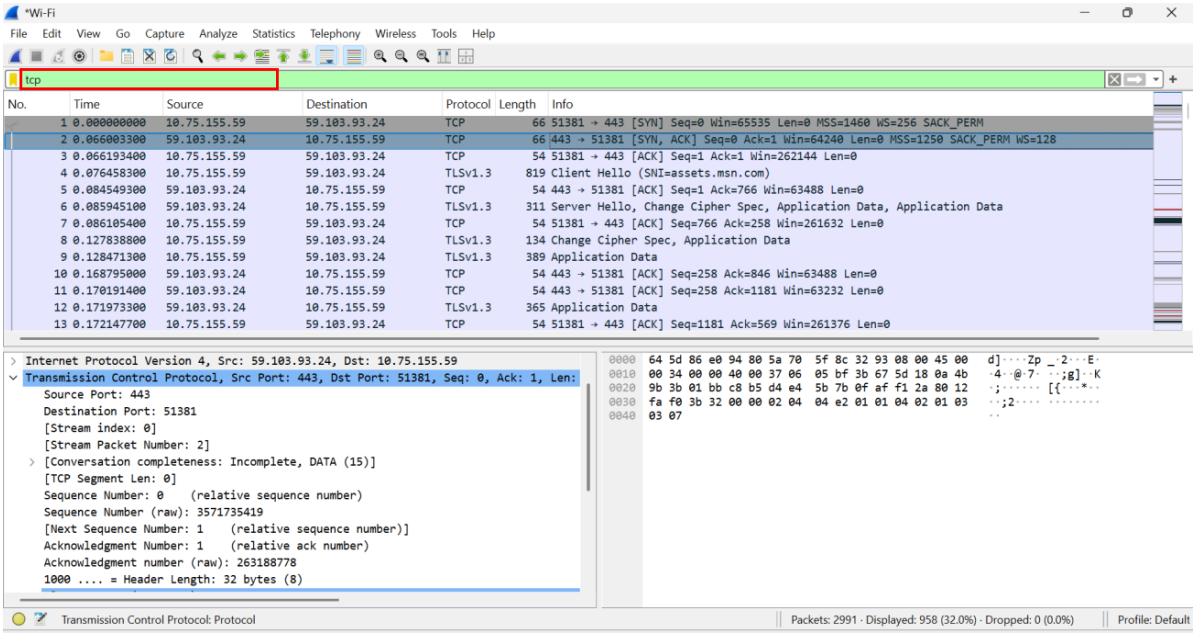


Figure 2: Display filter tcp applied to the capture

Defender view: Filtering by protocol is the first triage step. Instead of reading 2991 packets, the analyst reads the 958 that match the question being asked.

TCP retransmission

Purpose: To find a packet that was sent again. A retransmission happens when the sender does not get a reply in time. Wireshark detects this automatically and marks the packet.

Action: Keep the tcp filter applied and look for rows with a black background and the text [TCP Retransmission] in the Info column.

tcp

The same tcp filter as in the previous section. Wireshark colors problem packets black with red text.

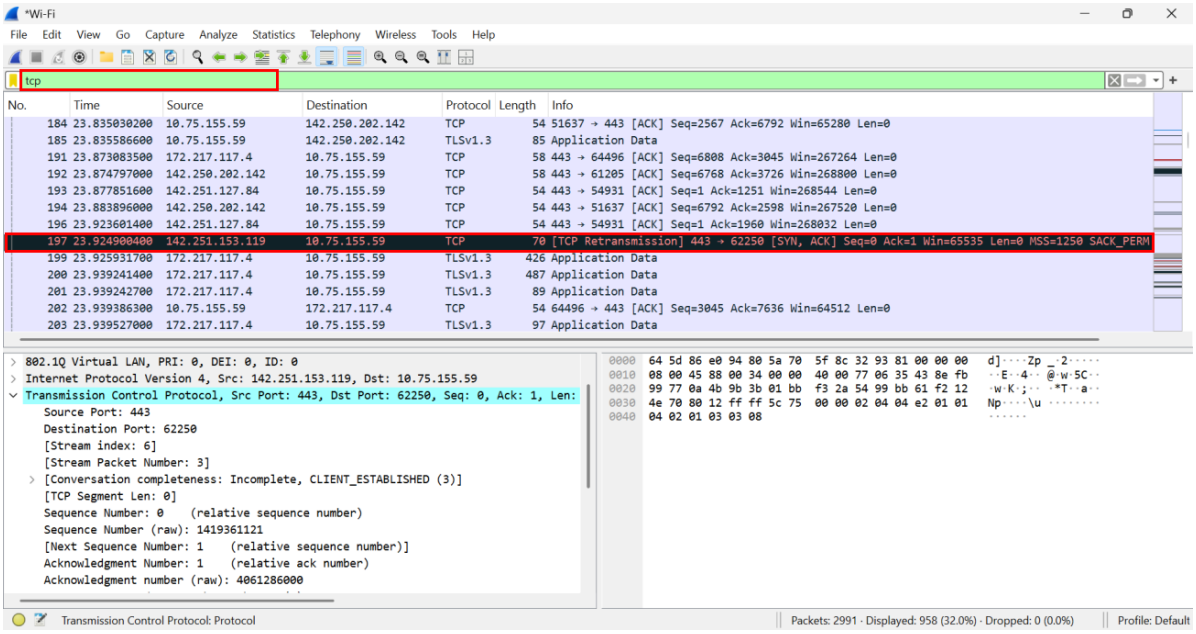


Figure 3: A packet marked as TCP Retransmission

Defender view: A few retransmissions are normal on Wi-Fi. A large number of them toward one destination can show a slow, blocked or unreachable service. The section on locating suspicious traffic shows an example of repeated retransmissions.

Display filter for UDP

Purpose: To show only UDP (User Datagram Protocol) packets. UDP has no handshake, so each packet is sent on its own. DNS and QUIC both use UDP.

Action: Type the filter below in the display filter bar and press Enter.

udp

udp: show only packets that use the UDP protocol.

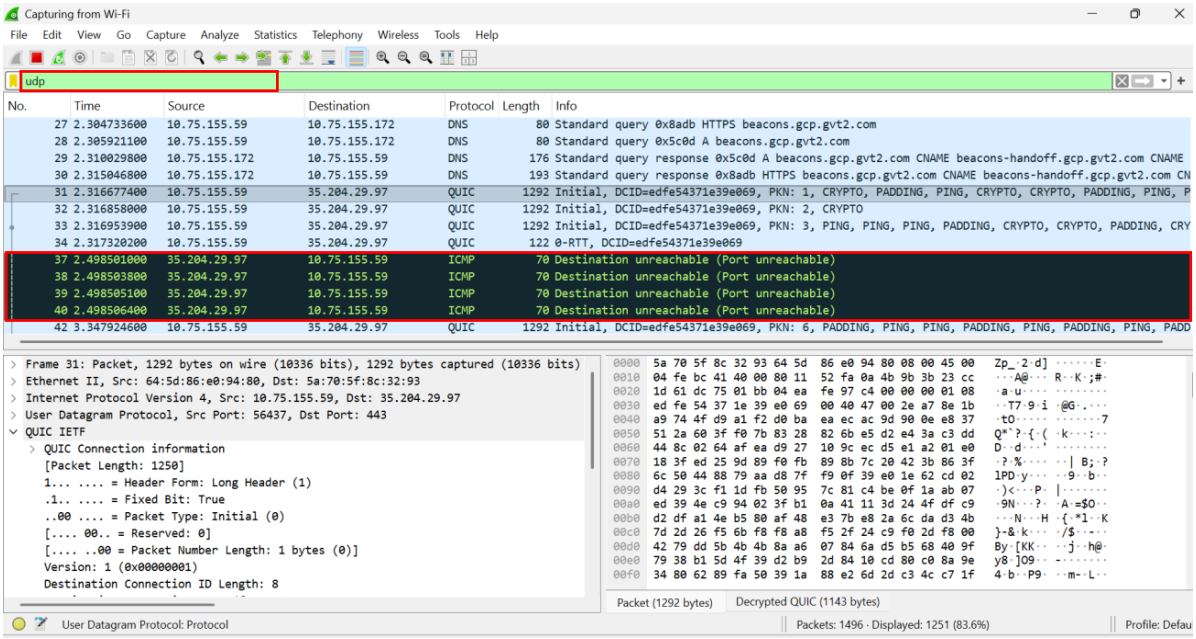


Figure 4: Display filter udp during a live capture, with ICMP port unreachable messages

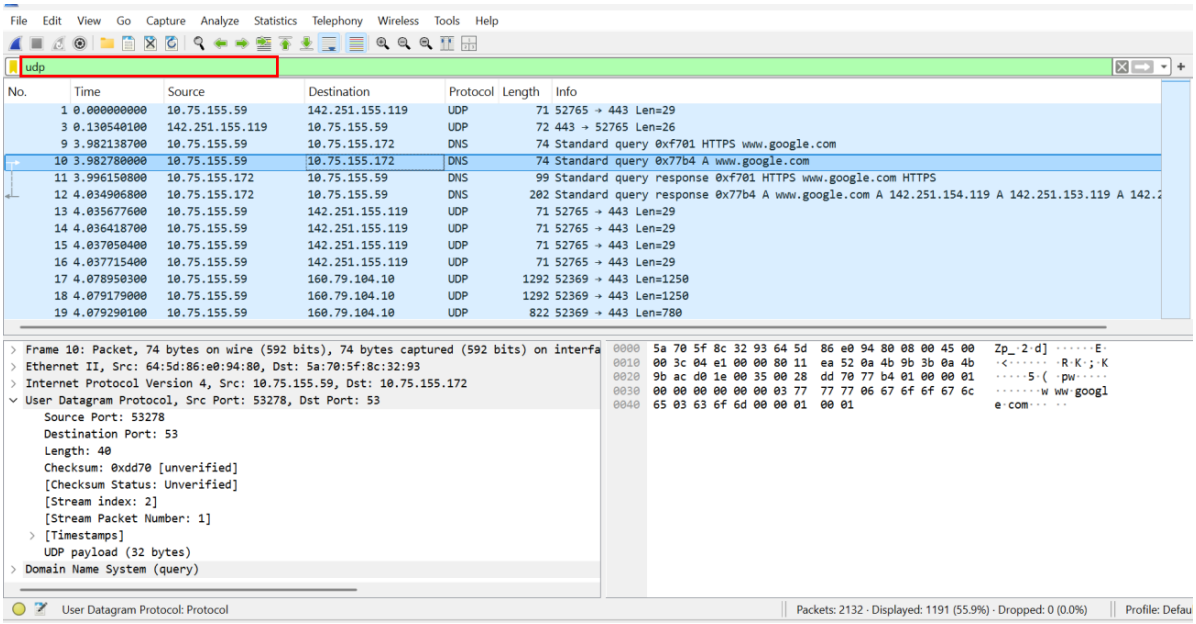


Figure 5: Display filter udp showing a DNS query and its UDP header

Defender view: UDP has no handshake, so it is easy to send and harder to track as a connection. Analysts watch UDP port 53 (DNS) closely because DNS shows which names a computer asked for.

Display filter for DNS

Purpose: To see only the DNS (Domain Name System) traffic. DNS turns a name such as www.google.com into an IP address.

Action: Type the filter below in the display filter bar and press Enter.

dns

dns: show only DNS queries and responses.

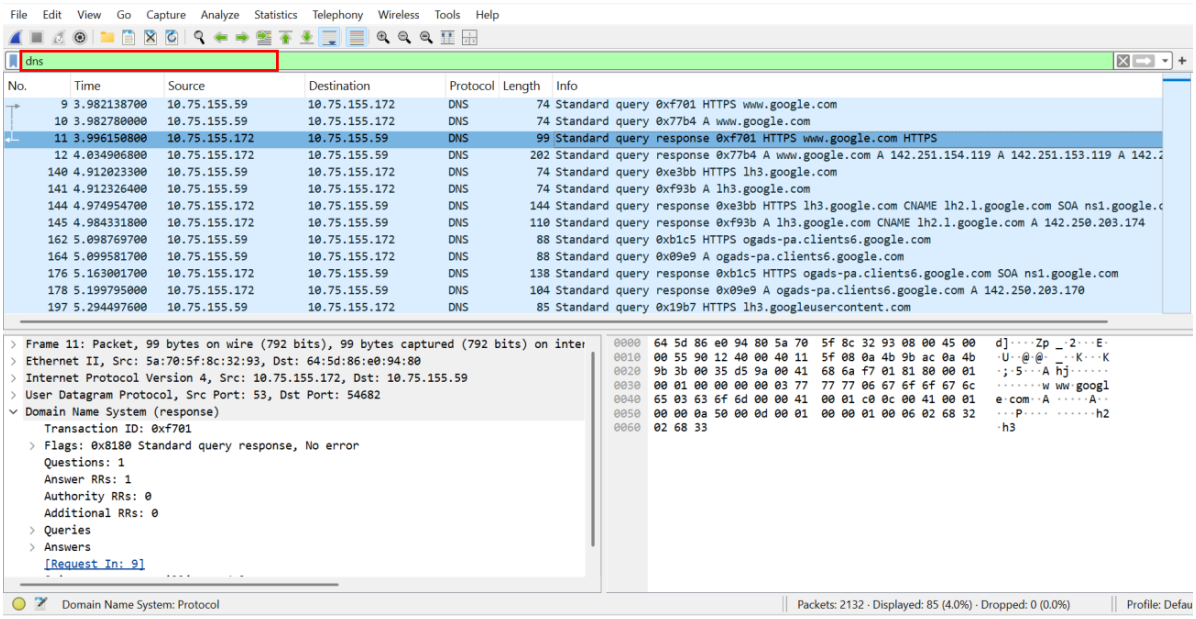


Figure 6: Display filter dns showing queries and responses

Defender view: DNS logs are one of the most useful data sources for a SOC. Analysts look for names that are very long, names that look random, and many failed lookups, because malware often uses DNS.

Display filter for ICMP

Purpose: To show only ICMP (Internet Control Message Protocol) packets. The ping command uses ICMP to test whether another computer can be reached.

Action: Type the filter below in the display filter bar and press Enter.

icmp

icmp: show only ICMP packets, such as ping requests and replies.

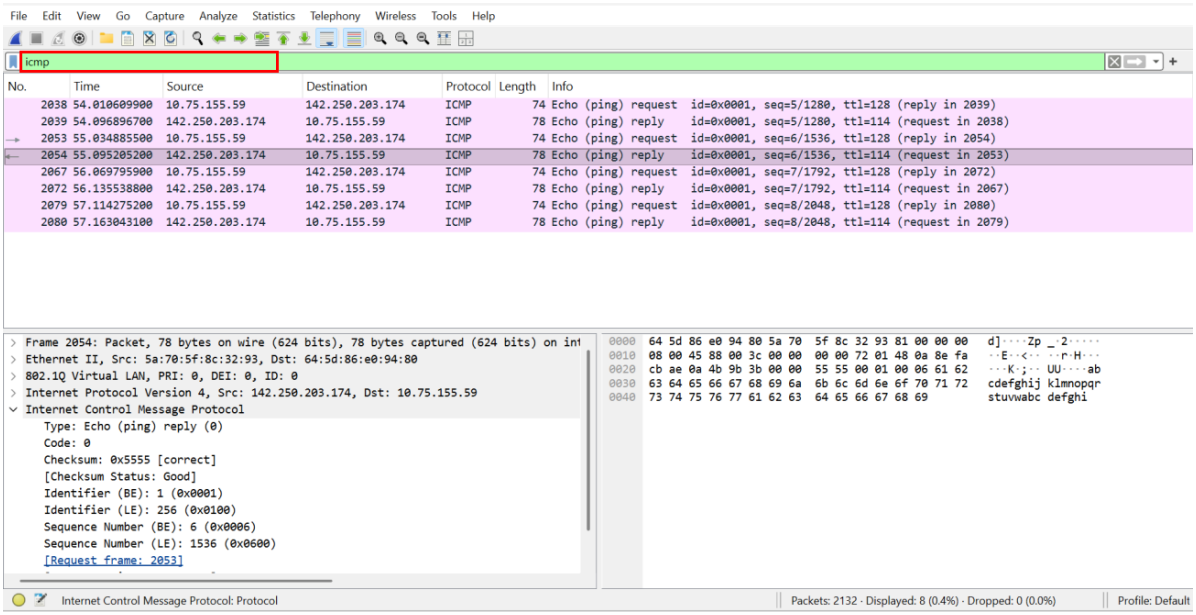


Figure 7: Display filter icmp showing ping requests and replies

Defender view: Ping is useful, but attackers also use it to find which computers are alive. Many echo requests from one source to many addresses is a pattern called a ping sweep.

Packet details and protocol layers

Purpose: To read one packet layer by layer. Each layer adds its own header, and the packet details pane lets us open every layer.

Action: Click a packet in the packet list, then click the arrow next to each layer in the packet details pane to expand it.

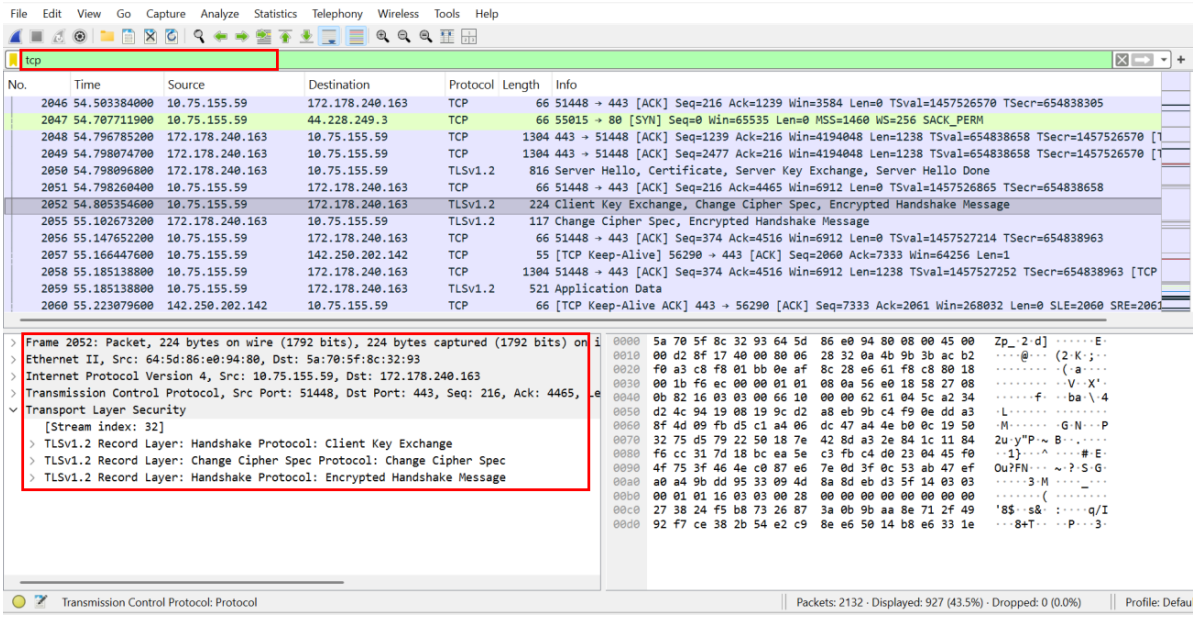


Figure 8: Protocol layers of a TLS packet in the packet details pane

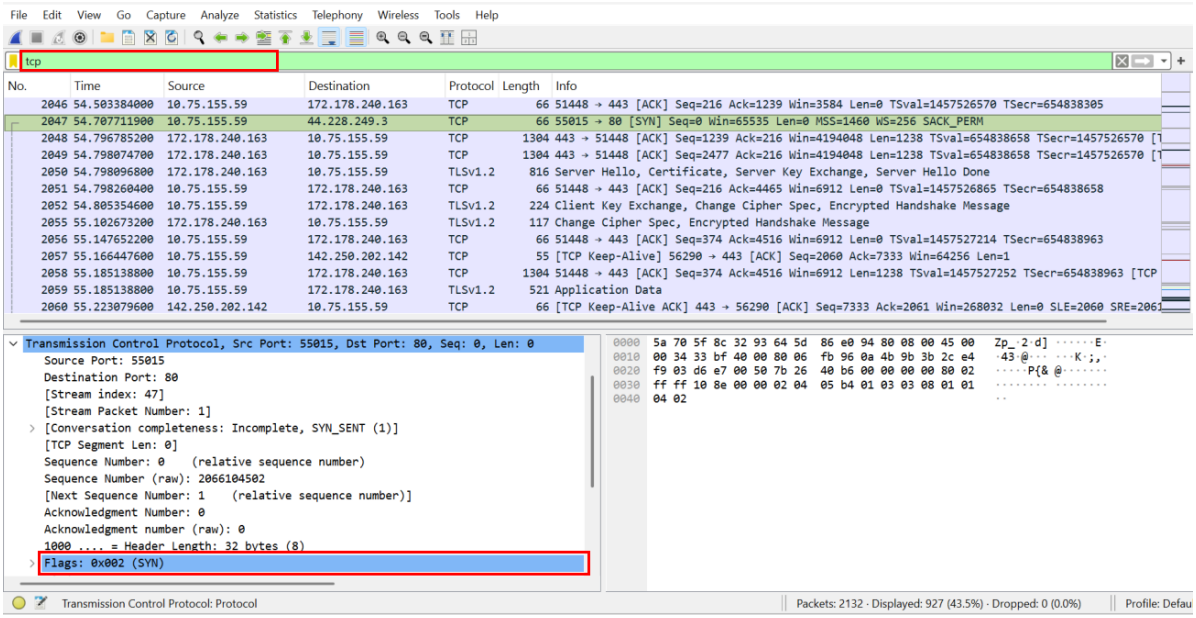


Figure 9: TCP flags of a SYN packet in the packet details pane

Defender view: Knowing the layers helps an analyst decide where a problem is. A wrong MAC address points to the local network, a wrong IP address points to routing, and a strange port or flag points to the application or to a scan.

Exporting packets

Purpose: To save a part of the capture into a new file, so that it can be shared or studied later without the rest of the capture.

Action: Open File, then Export Specified Packets. Choose the file name, the file type and the packet range, then press Save.

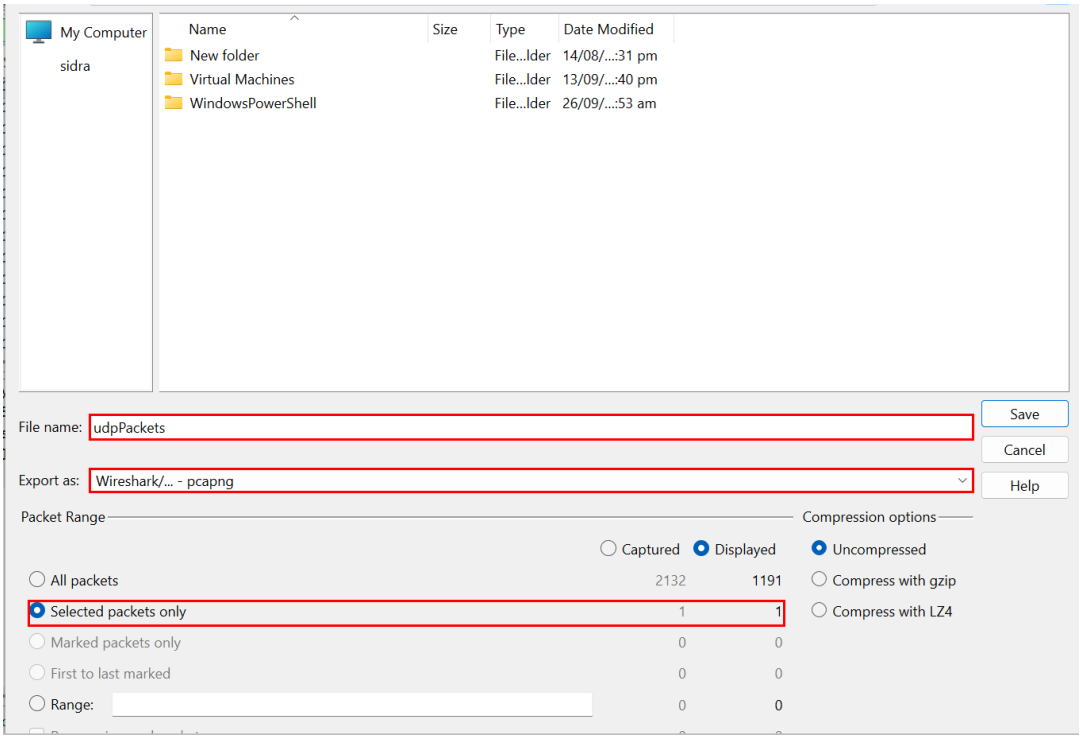


Figure 10: Export Specified Packets window

Defender view: Capture files are evidence. Keep the original capture unchanged, and share only filtered copies. Remove sensitive data before sharing a file outside the team.

Follow TCP Stream

Purpose: To join all packets of one TCP conversation and show the data in the order it was sent.

Action: Right-click a TCP packet and choose Follow, then TCP Stream.

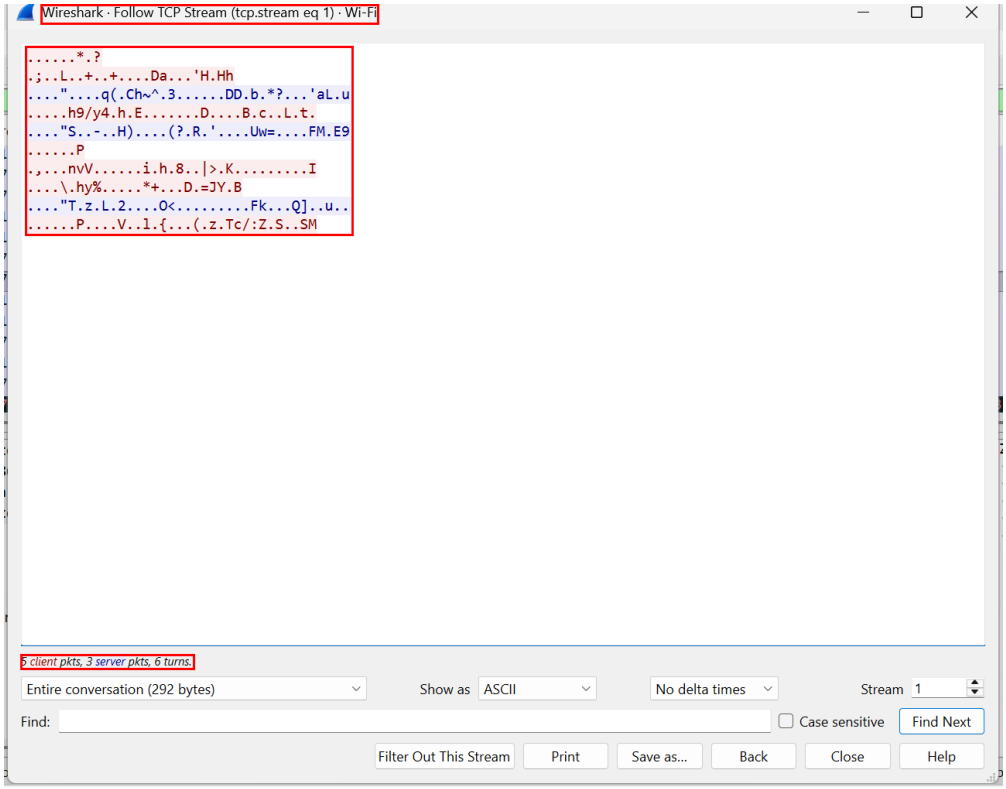


Figure 11: Follow TCP Stream window for stream 1

Defender view: For protocols that are not encrypted, such as HTTP, FTP and Telnet, this view can show commands and passwords in clear text. This is why those protocols are risky. For encrypted traffic, nothing readable appears, so the analyst studies the metadata instead.

Expert information

Purpose: To let Wireshark list the problems and unusual events it found in the capture, so that the analyst does not have to read every packet.

Action: Open the Analyze menu and choose Expert Information.

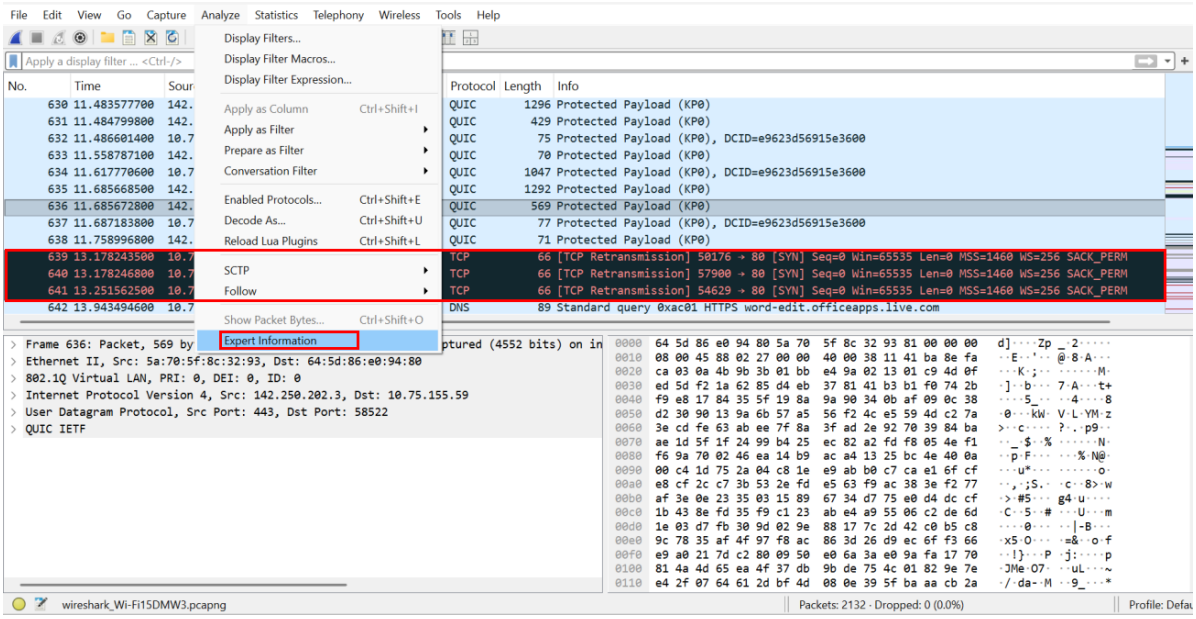


Figure 12: Opening Expert Information from the Analyze menu

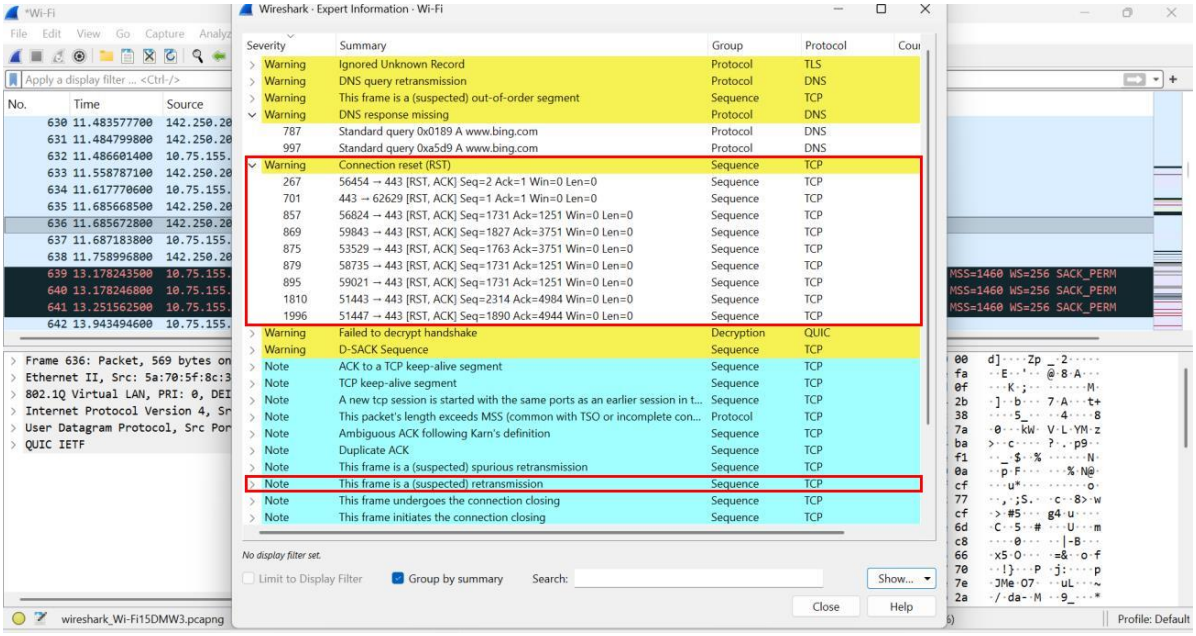


Figure 13: Expert Information window

Defender view: Expert Information is a quick triage list. Many resets, retransmissions or missing DNS responses can point to network problems, blocked connections or scanning. The analyst starts here, then goes to the packets behind each item.

Conversations

Purpose: To see which pairs of addresses talked to each other, how many packets they exchanged and for how long.

Action: Open the Statistics menu and choose Conversations.

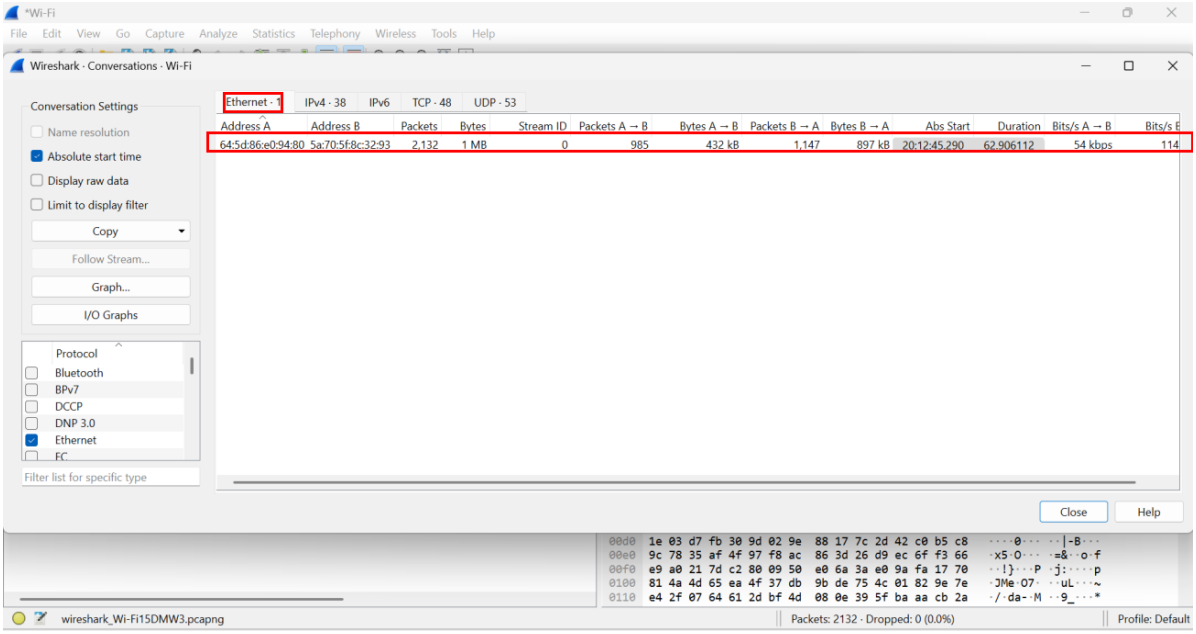


Figure 14: Conversations window (Ethernet tab)

Defender view: Conversations answer the question "who talked to whom, and how much". An analyst sorts by bytes. A single computer that sends a very large amount of data to one outside address may be a sign of data theft, so it needs a closer look.

Endpoints

Purpose: To list each single device (endpoint) in the capture, with the packets and bytes it sent and received.

Action: Open the Statistics menu and choose Endpoints.

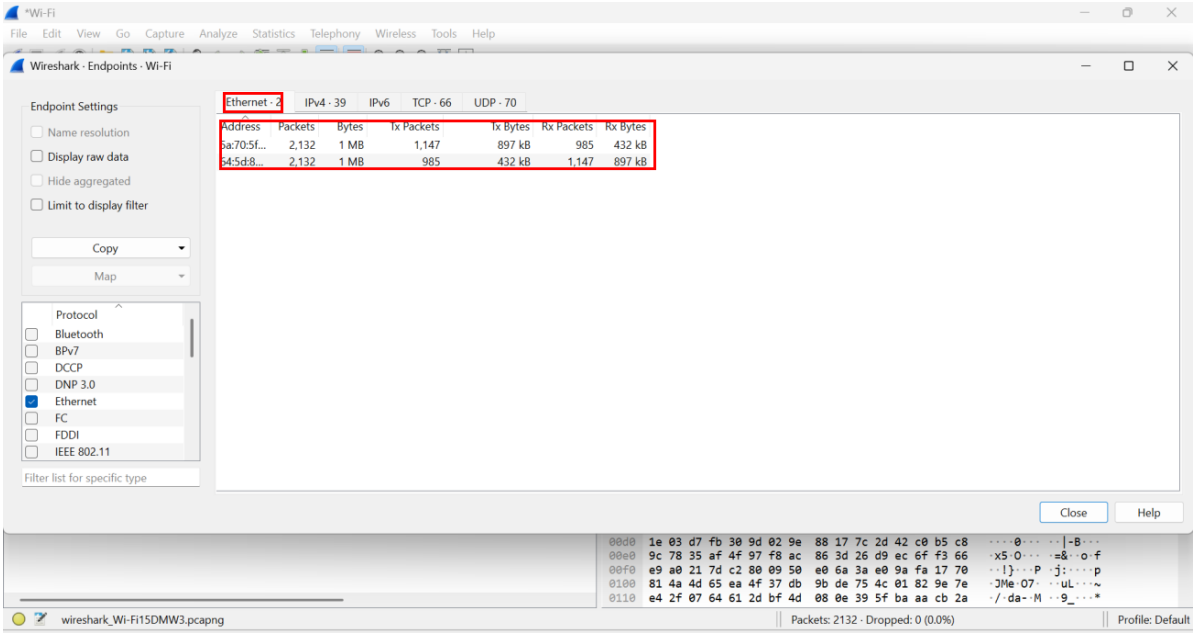


Figure 15: Endpoints window (Ethernet tab)

Defender view: Endpoints show the "top talkers". A device with an unexpectedly high number of packets or bytes, or an unknown IPv4 address in the list, is a candidate for investigation.

Locating suspicious traffic

Purpose: To find new connection attempts and look for unusual patterns in them, such as the same attempt repeated again and again.

Action: Type the filter below in the display filter bar and press Enter.

tcp.flags.syn == 1 && tcp.flags.ack == 0

tcp.flags.syn == 1: the SYN flag is switched on. &&: and. tcp.flags.ack == 0: the ACK flag is switched off. Together they show only the first packet of each new TCP connection attempt.

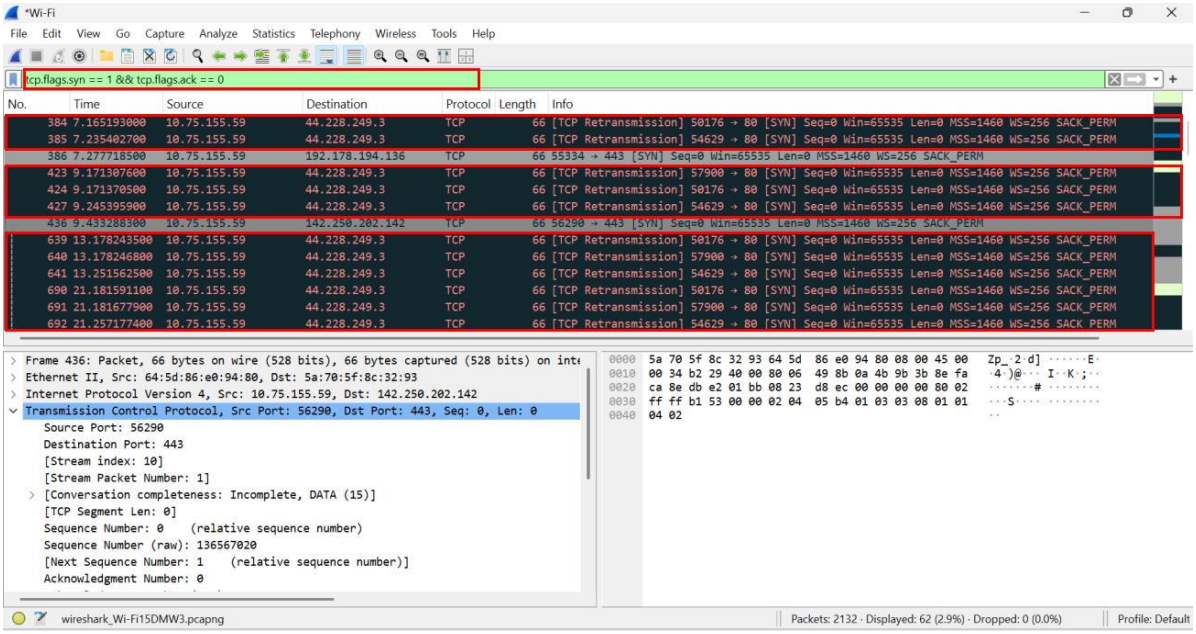


Figure 16: Repeated SYN retransmissions found with a SYN filter

Defender view: This is the same filter that helps to find port scans, because a scan sends many SYN packets. Here the pattern is different: the same destination and the same port are tried again and again, so it looks like failed connection attempts and not a scan. The next step for an analyst is to find which program owns the connection, for example with the netstat -ano command on Windows, and to check whether the destination is expected.

Name resolution

Purpose: To show names instead of numbers. Wireshark can turn MAC addresses, IP addresses and port numbers into names that are easier to read.

Action: Open the View menu and choose Name Resolution. Then choose one of the options.

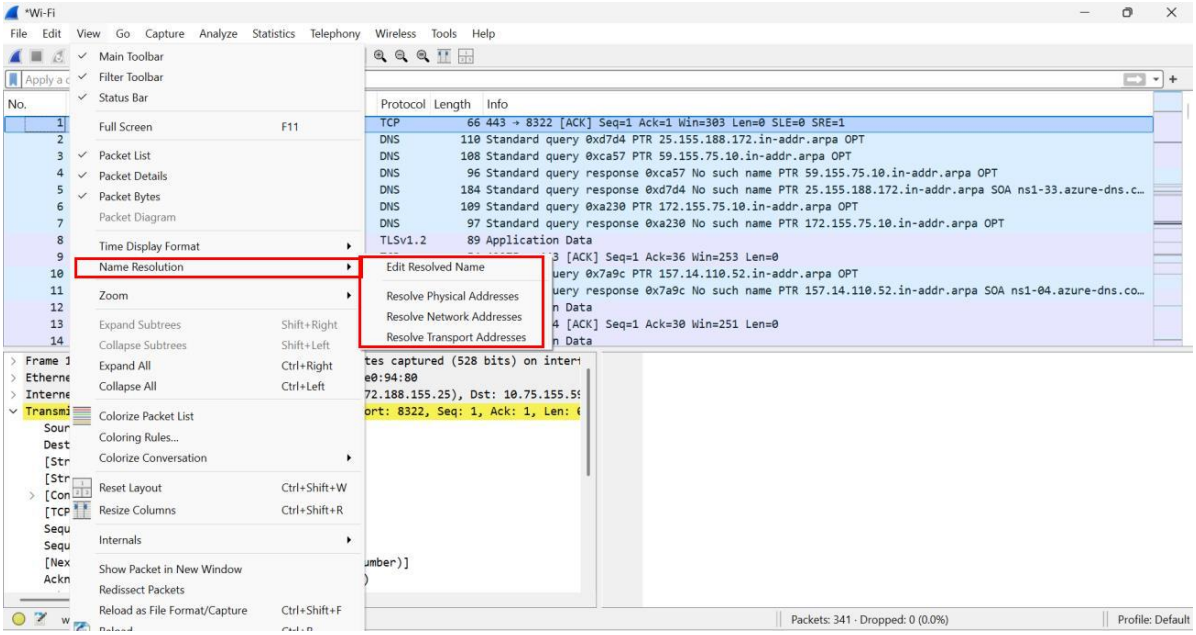


Figure 17: Name Resolution options in the View menu

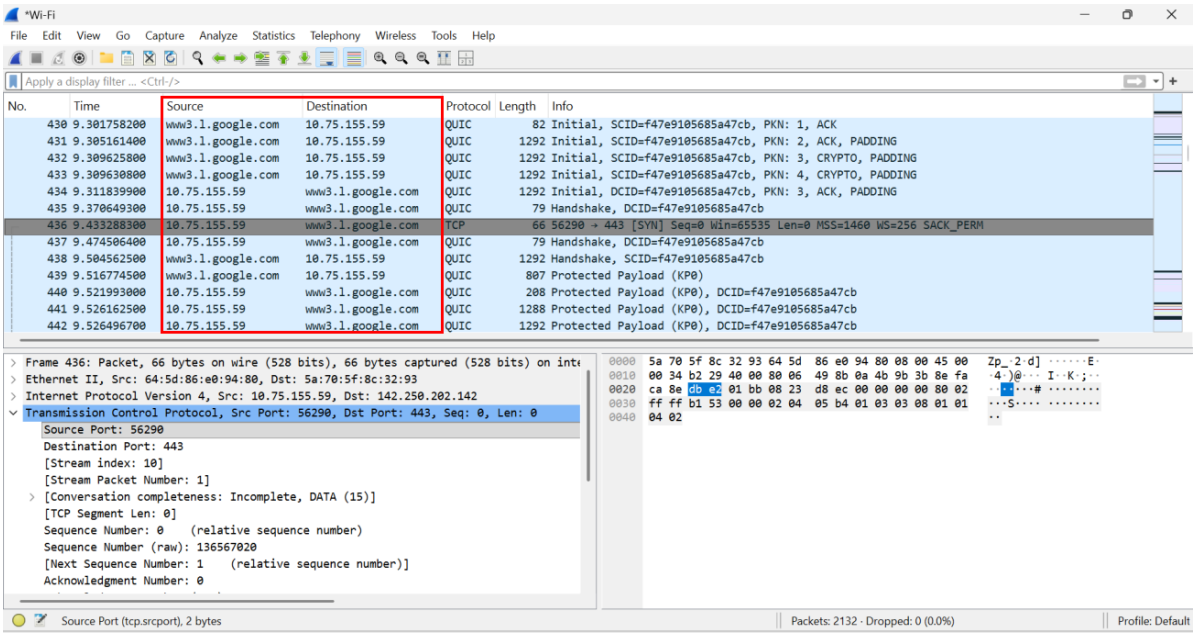


Figure 18: Host names shown in the Source and Destination columns

Defender view: Names make a capture easier to read. But resolving names makes Wireshark send DNS queries of its own, which tells the DNS server which addresses are being studied.

When an analyst investigates a suspicious address, name resolution is often turned off for this reason.

Time display format

Purpose: To change how the Time column is shown, so that packets can be matched with other records.

Action: Open the View menu and choose Time Display Format. The options include seconds since the beginning of the capture, date and time of day, and seconds since the previous displayed packet (the time delta).

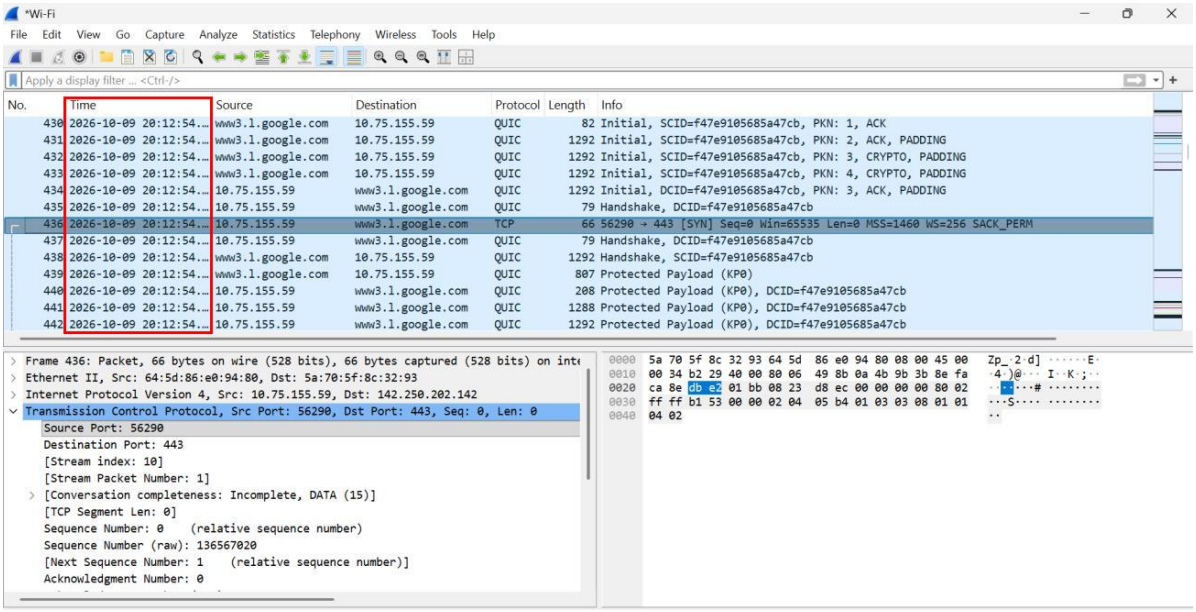


Figure 19: Time column shown as date and time of day

Defender view: The date and time of day lets an analyst match a packet with a firewall log or a SIEM alert from the same moment. Time deltas help to find gaps and regular intervals, which can show automatic behavior such as beaconing.

PART B: NMAP PRACTICAL WITH ZENMAP

In this part, Nmap was run through Zenmap. The Command box of Zenmap shows the exact Nmap command for every scan. Scans were run only against this computer, a web server whose owner gave permission, the university website, and scanme.nmap.org, which is provided for practice.

Intense Scan with Service Detection

Purpose: To find which ports are open on this computer and which service is behind each of them. The target is the address 192.168.100.1.

Action: Enter the target in Zenmap and choose the profile "Intense scan". Zenmap builds and runs this command:

nmap -T4 -A -v 192.168.100.1

- nmap: the scanning program.
- -T4: timing template 4. It makes the scan faster. The scale goes from 0 (slowest) to 5 (fastest).
- -A: aggressive scan. It turns on OS detection, version detection, default scripts and traceroute together.
- -v: verbose. Nmap prints each stage of the scan while it works.
- 192.168.100.1: the target address.

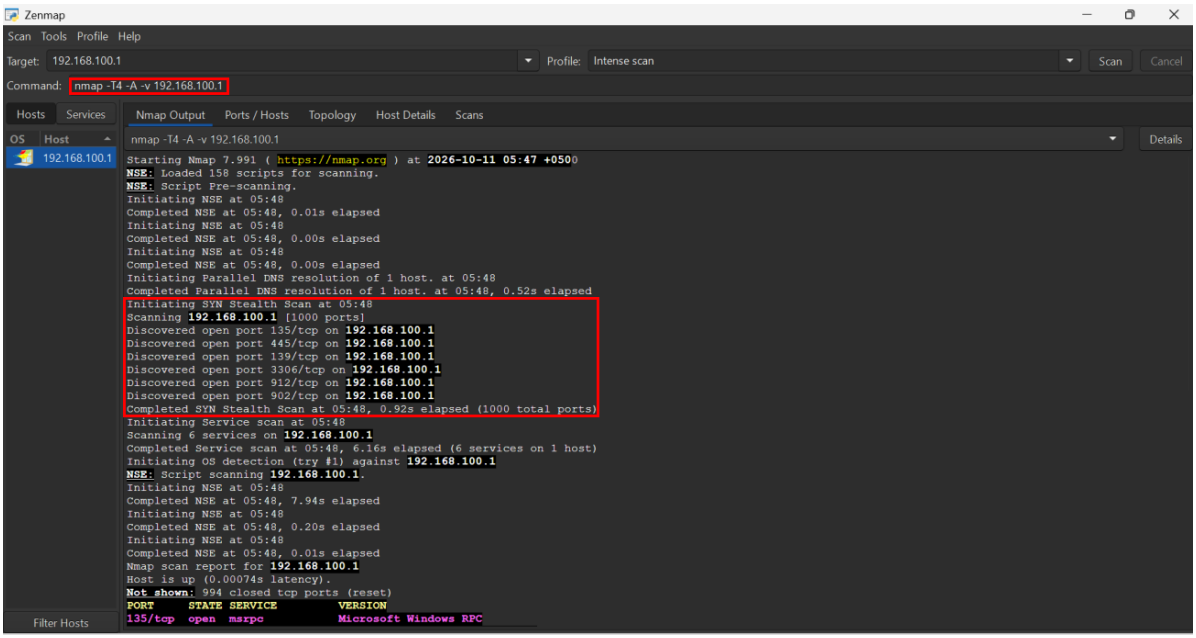


Figure 20: Zenmap intense scan output

Result: Nmap used a SYN scan (it sends a SYN packet and does not finish the handshake) and checked 1000 ports in 0.92 seconds. Six ports are open: 135 (msrpc, Microsoft RPC, which lets

Windows programs talk to each other), 445 (SMB, Windows file sharing), 139 (NetBIOS, older Windows file and printer sharing), 3306 (MySQL database server), and 912 and 902 (VMware authentication service). The other 994 ports are closed. The screenshot ends at the first row of the port table, and only port 135 shows its service details there, so the purposes of the other ports are the usual uses of those port numbers.

Defender view: A defender compares the list of open ports with the list that should be open. Every extra open port is a possible way in. A database port such as 3306 on a normal computer is worth checking. A scan from the same computer may also not show exactly what other devices on the network can reach.

Fast Scan of a Web Server

Purpose: To scan the web server www.douglas.stebila.ca, whose owner gave permission for this scan.

Action: Type the command in the Command box of Zenmap and run it:

nmap -T4 -F www.douglas.stebila.ca

- -T4: faster timing.
- -F: fast scan. It checks only the 100 most common ports and not the default 1000.
- www.douglas.stebila.ca: the target. Nmap first looks up its IP address with DNS.

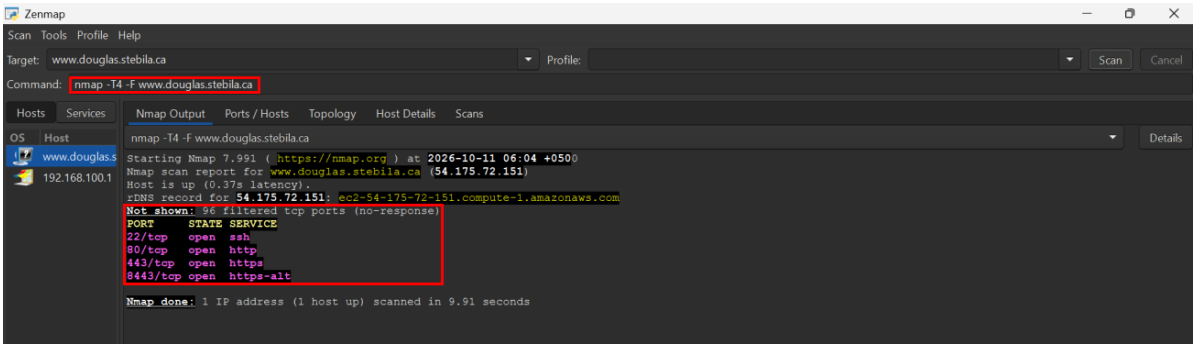


Figure 21: Fast scan of www.douglas.stebila.ca

Result: Four ports are open: 22 (ssh, encrypted remote login), 80 (http, web pages without encryption), 443 (https, web pages with encryption) and 8443 (https-alt, an alternative port for HTTPS web services). The other 96 of the 100 ports checked are filtered, which means the probes got no answer and a firewall probably dropped them. Nmap also found the reverse DNS name ec2-54-175-72-151.compute-1.amazonaws.com, which is an Amazon cloud server name.

Defender view: A web server needs ports 80 and 443. An SSH port that is open to the internet receives constant login attempts, so it should use keys, restrict who can reach it, and be monitored for many failed logins.

Fast Scan of a Second Web Server

Purpose: To scan the university website www.iiu.edu.pk.

Action: Enter the target in Zenmap and choose the profile "Quick scan". Zenmap builds and runs this command:

nmap -T4 -F www.iiu.edu.pk

- -T4: faster timing.
- -F: fast scan of the 100 most common ports.
- www.iiu.edu.pk: the target.

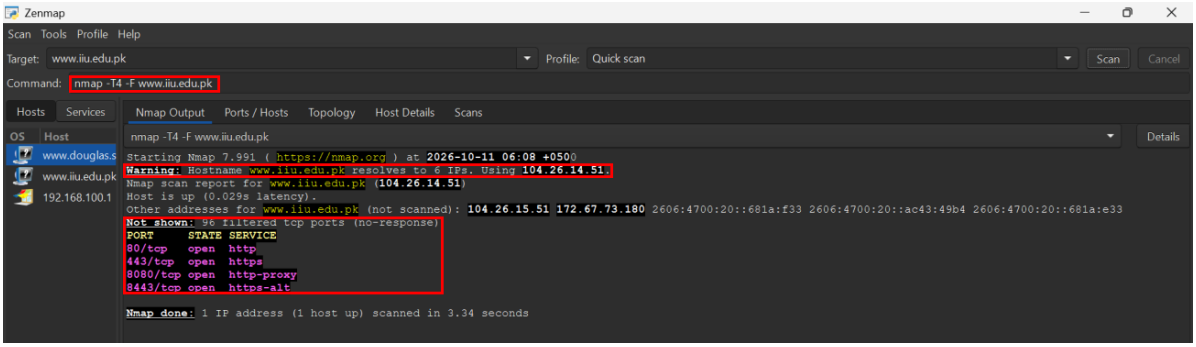


Figure 22: Fast scan of www.iiu.edu.pk

Result: Four ports are open: 80 (http, web pages without encryption), 443 (https, web pages with encryption), 8080 (http-proxy, an alternative port for HTTP web services or proxies) and 8443 (https-alt, an alternative port for HTTPS web services). The other 96 ports are filtered. Nmap warned that the host name resolves to 6 IP addresses and scanned only one of them, so the result describes that one address. Several addresses for one name are common when a site uses a content delivery or protection service, but the screenshot does not confirm this.

Defender view: Only web-type ports are open here, which is what a public website normally needs. Fast scans such as this one are light, but a scan of a system should only be done with permission.

Fast Scan of a Practice Server

Purpose: To scan a server that the Nmap project provides for people to practice scanning.

Action: Enter the target in Zenmap and choose the profile "Quick scan". Zenmap builds and runs this command:

nmap -T4 -F scanme.nmap.org

- -T4: faster timing.
- -F: fast scan of the 100 most common ports.
- scanme.nmap.org: the practice target.

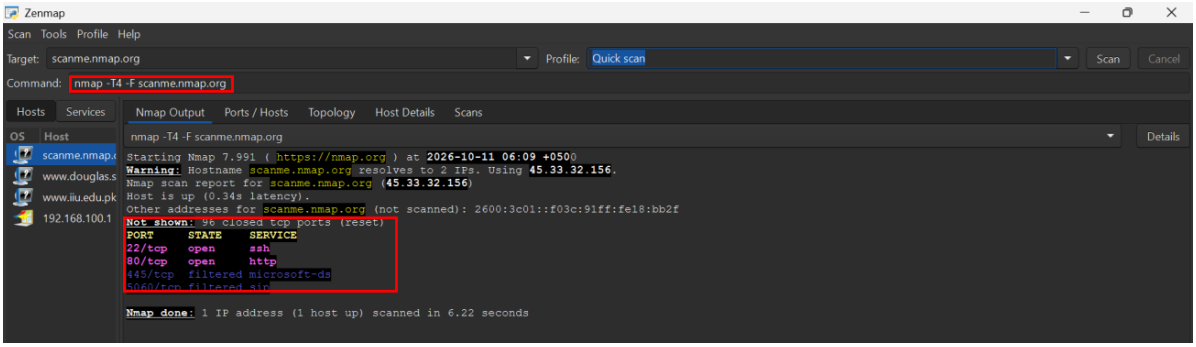


Figure 23: Fast scan of scanme.nmap.org

Result: Two ports are open: 22 (ssh, encrypted remote login) and 80 (http, web pages without encryption). Two ports are filtered: 445 (microsoft-ds, SMB file sharing) and 5060 (sip, call setup for internet phone services). The other 96 ports are closed, which means the host answered with a reset. Nmap used the IPv4 address and did not scan the IPv6 address.

Open, closed and filtered compared:

- Open: a program is listening and answers the probe.
- Closed: the host answers with a reset, so no program is listening on that port. Figure 23 shows 96 closed ports.
- Filtered: no answer comes back, so a firewall probably drops the probe. Figures 21 and 22 each show 96 filtered ports.
- So the three servers treat unused ports differently. www.douglas.stebila.ca and the IIUI website stay silent, and scanme.nmap.org answers with resets.

Defender view: The difference between closed and filtered matters to a defender. A closed port answers with a reset, so it tells the scanner that the host is there. A filtered port gives no answer at all, which slows a scanner and reveals less.

CONCLUSION

This practical showed how two common tools work together in security analysis. Wireshark shows what is really sent over the network, packet by packet, and its filters, statistics and Expert Information make a large capture easy to study. Nmap shows what a system offers to the network, and its states open, closed and filtered show how a firewall treats each port.

Key findings

- Filtering a capture by protocol reduced it to the packets that matter, for example 85 DNS packets out of 2132 and 8 ICMP packets out of 2132.
- Expert Information listed nine connection resets and several retransmissions without reading each packet.
- The SYN filter showed repeated retransmitted SYN packets from the computer to one destination on port 80.
- The intense scan of this computer found six open ports, including 445 (SMB) and 3306 (MySQL).
- The three fast scans showed three different results: open web ports with many filtered ports, and one server that answers unused ports with resets.

Limits of this work

- The packet counts differ between some figures because more than one capture was made.
- A scan from the same computer may not show exactly what other devices on the network can reach.
- A fast scan checks only 100 ports, so ports outside that list were not tested.